

Chapter 3. GROUPS

3.3 Cyclic Groups



Today we will focus our attention on **CYCLIC GROUPS**.
We will

- Introduce a more formal definition.
- See several examples.
- Look at their Cayley tables.

On the definition of a cyclic group

Recall that **cyclic groups** are groups whose elements are all powers (or multiples) of a fixed element, called the **generator of the group**.

The cyclic group of order n is

$$C_n = \{1, a, a^2, \dots, a^{n-1}\}, \text{ where } a^n = 1.$$

We write $C_n = \langle a \rangle$ and call a a **generator of C_n** . By $|C_n| = n$, we mean that the elements $1, a, a^2, \dots, a^{n-1}$ are **all distinct elements** of C_n .

The **Cayley table** of C_n is determined completely by the exponent laws and the condition $a^n = 1$.

C_n	1	a	a^2	$\dots$	a^{n-2}	a^{n-1}
1	1	a	a^2	$\dots$	a^{n-2}	a^{n-1}
a	a	a^2	a^3	$\dots$	a^{n-1}	1
a^2	a^2	a^3	a^4	$\dots$	1	a
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
a^{n-2}	a^{n-2}	a^{n-1}	1	$\dots$	a^{n-4}	a^{n-3}
a^{n-1}	a^{n-1}	1	a	$\dots$	a^{n-3}	a^{n-2}

REMARK: When computing the Cayley table of C_n we will have to deal with elements of the form a^k , for $k > n$.

What do we do?

$$C_5 = \{1, a, a^2, a^3, a^4\} \text{ and } a^5 = 1; \quad a^{108} = ?$$

① Division algorithm: $108 = 5 \cdot 21 + 3$

$$\begin{aligned} \text{② Exponent laws: } a^{108} &= a^{5 \cdot 21 + 3} = \underbrace{(a^5)^{21}}_{a^5=1} \cdot a^3 \stackrel{\text{②}}{=} a^3 \\ &= \underline{a^3} \end{aligned}$$

Definition 3.22. Let g be an element of a group G , the subset of G consisting of all the powers of g , namely, $\langle g \rangle = \{g^k \mid k \in \mathbb{Z}\}$ is a **subgroup of G** (this follows from the exponent laws, please fill in the details), called the **cyclic subgroup of G generated by g** . Moreover, $\langle g \rangle$ is the **smallest subgroup of G containing the element g** .

We will say that G is a **cyclic group** if there exists $g \in G$ such that $G = \langle g \rangle$. Notice that if g has order n , then G is a **cyclic group of order n** ; if g has order infinity, we say that G is a **cyclic group of order infinity**.

Remark 3.23. If the operation in the group G is **written additively**, the **cyclic subgroup generated by g** consists on the multiples of g :

$$\langle g \rangle = \{kg \mid k \in \mathbb{Z}\} = \mathbb{Z}g.$$

Remember that in additive notation “powers” are replaced by “multiples”.

Examples 3.24.

1. The cyclic subgroup generated by $\sigma \in S_3$ is $\langle \sigma \rangle = \{\varepsilon, \sigma, \sigma^2\}$.
2. The cyclic subgroup generated by $2 \in \mathbb{Z}$ consists on the even numbers.
3. The group $(\mathbb{Z}, +)$ is cyclic and 1 and -1 are the only generators.

If $k \in \mathbb{Z}$ we have $k = k \cdot 1 \in \langle 1 \rangle$ and $k = (-k) \cdot (-1) \in \langle -1 \rangle$. Thus $\mathbb{Z} = \langle 1 \rangle = \langle -1 \rangle$. Moreover, if n is an integer different from 1 and -1 then $n\mathbb{Z} \neq \mathbb{Z}$ since, for example, $n + 1 \notin n\mathbb{Z}$.

4. The group $(\mathbb{Z}_n, +)$ is cyclic with generator $\bar{1}$ and order n .

Recall that $\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$. Given $\bar{k} \in \mathbb{Z}_n$ we have that $\bar{k} = k\bar{1}$ and so $\bar{k} \in \langle \bar{1} \rangle$. It follows that $\mathbb{Z}_n = \langle \bar{1} \rangle$, as desired.

Let us compare the additive cyclic group $\mathbb{Z}_n$ with the multiplicative cyclic group C_n .

$$C_n = \langle a \rangle = \{a^0, a^1, \dots, a^{n-1}\} \quad (a^n = 1) \quad (a^0 = 1)$$

$\parallel$


$$\mathbb{Z}_n = \langle \bar{1} \rangle = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$$

C_n and $\mathbb{Z}_n$ are isomorphic!

6. The (multiplicative) group $\mathbb{Z}_9^* = \{\bar{1}, \bar{2}, \bar{4}, \bar{5}, \bar{7}, \bar{8}\}$ consisting on the units of $\mathbb{Z}_9$ is a cyclic group. In fact, $\mathbb{Z}_9^* = \langle \bar{2} \rangle$ because

Goal: Every element of $\mathbb{Z}_9^*$ is a power of $\bar{2}$:

$$\bar{2}^0 = \bar{1}, \quad \bar{2}^1 = \bar{2}, \quad \bar{2}^2 = \bar{2} \cdot \bar{2} = \overline{2 \cdot 2} = \bar{4}$$

$$\bar{2}^3 = \bar{2} \cdot \bar{2} \cdot \bar{2} = \overline{2^3} = \bar{8}$$

$$\bar{2}^4 = \overline{2^4} = \overline{16} = \bar{7}$$

$$\bar{2}^5 = \overline{14} = \bar{5} \quad \text{and} \quad \bar{2}^6 = \bar{9}$$

$$\bar{2}^5 = \bar{2}^4 \cdot \bar{2} = \bar{7} \cdot \bar{2} = \overline{14} = \dots$$

7. The (multiplicative) group $\mathbb{Z}_8^* = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ is not cyclic. Recall that $\mathbb{Z}_8^*$ consists on the units $\mathbb{Z}_8$, that is, elements of $\mathbb{Z}_8 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}\}$, having a multiplicative inverse.

To show that $\mathbb{Z}_8^*$ is not cyclic we need to prove $\langle a \rangle \neq \mathbb{Z}_8^*$ for all $a \in \mathbb{Z}_8^*$.

In fact:

$$\langle \bar{1} \rangle = \{\bar{1}\}$$

$$\langle \bar{3} \rangle = \{\bar{1}, \bar{3}\}$$

$$\bar{3}^0 = \bar{1}$$

$$\bar{3}^1 = \bar{3}$$

$$\bar{3}^2 = \bar{9} = \bar{1}$$

$$\langle \bar{5} \rangle = \{\bar{1}, \bar{5}\}$$

$$\bar{5}^0 = \bar{1}$$

$$\bar{5}^1 = \bar{5}$$

$$\bar{5}^2 = \bar{25} = \bar{1}$$

$$\langle \bar{7} \rangle = \{\bar{1}, \bar{7}\}$$

$$\bar{7}^0 = \bar{1}$$

$$\bar{7}^1 = \bar{7}$$

$$\bar{7}^2 = \bar{49} = \bar{1}$$